Regret Bound for LSVI-UCB on Linear MDPs

1 Preliminaries

Notation. For a vector $v \in \mathbb{R}^d$ and a positive-definite matrix $A \succ 0$, write $\|v\|_A := \sqrt{v^\top A v}$. For a function $f\colon \mathcal{S} \to \mathbb{R}$ and a kernel $P_h(\cdot \mid s, a)$, write $[P_h f](s,a) := \mathbb{E}_{s' \sim P_h(\cdot|s,a)}[f(s')]$. Throughout, $c, C, C_1, C_2, \ldots$ denote universal positive constants whose values may change from line to line and depend only on quantities specified in each lemma's hypotheses.

MDP setting. We consider a finite-horizon, episodic Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, H, \{P_h\}_{h=1}^H, \{r_h\}_{h=1}^H)$, where $\mathcal{S}$ is a (possibly infinite) state space, $\mathcal{A}$ is a finite action space, $H \in \mathbb{N}$ is the horizon, $P_h(\cdot \mid s, a)$ is the transition kernel at step h, and $r_h\colon \mathcal{S} \times \mathcal{A} \to [0,1]$ is the deterministic reward. A policy is a sequence $\pi = (\pi_h)_{h=1}^H$ with $\pi_h\colon \mathcal{S} \to \mathcal{A}$. For each policy π, define value and action-value functions

$$V_h^\pi(s) := \mathbb{E}^\pi\Big[\textstyle\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'}) \,\Big|\, s_h = s\Big], \quad Q_h^\pi(s,a) := \mathbb{E}^\pi\Big[\textstyle\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'}) \,\Big|\, s_h = s,\, a_h = a\Big].$$

Let $V_h^*(s) := \max_\pi V_h^\pi(s)$ and $Q_h^*(s,a) := \max_\pi Q_h^\pi(s,a)$ denote the optimal value functions, with $V_{H+1}^\pi \equiv 0$. The Bellman equation gives $Q_h^*(s,a) = r_h(s,a) + [P_h V_{h+1}^*](s,a)$ and $V_h^*(s) = \max_a Q_h^*(s,a)$.

An algorithm interacts with $\mathcal{M}$ over K episodes. At the start of episode k, it receives a fixed initial state s_1^k and follows a policy π^k, yielding a trajectory $(s_h^k, a_h^k)_{h=1}^H$ with $s_{h+1}^k \sim P_h(\cdot \mid s_h^k, a_h^k)$. The cumulative regret over K episodes is

$$\mathrm{Regret}(K) \;:=\; \sum_{k=1}^K \big[V_1^*(s_1^k) - V_1^{\pi^k}(s_1^k)\big].$$

Write $T := KH$ for the total number of step-level interactions.

Assumption 1.1 (Linear MDP)**.** There exists a known feature map $\phi\colon \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$ such that:

(i) (Linear transitions.) For each $h \in [H]$ there exists a signed vector measure $\mu_h = (\mu_h^{(1)}, \ldots, \mu_h^{(d)})$ on $\mathcal{S}$ such that for every $(s,a) \in \mathcal{S} \times \mathcal{A}$,

$$P_h(\cdot \mid s, a) \;=\; \langle \phi(s,a), \mu_h(\cdot) \rangle, \quad \text{with} \quad \|\mu_h(\mathcal{S})\|_2 \le \sqrt{d}.$$

(ii) (Linear rewards.) For each $h \in [H]$ there exists $\theta_h \in \mathbb{R}^d$ with $\|\theta_h\|_2 \le \sqrt{d}$ and $r_h(s,a) = \langle \phi(s,a), \theta_h \rangle$.

(iii) (Bounded features.) $\|\phi(s,a)\|_2 \le 1$ for all $(s,a) \in \mathcal{S} \times \mathcal{A}$.

Remark 1.2. Assumption 1.1 immediately implies that for any function $f\colon \mathcal{S} \to \mathbb{R}$, the conditional expectation $[P_h f](s,a) = \big\langle \phi(s,a), \int_{\mathcal{S}} f(s')\,\mathrm{d}\mu_h(s') \big\rangle$ is itself linear in $\phi(s,a)$. Combined with the linear reward, this means Q_h^π is linear in $\phi(s,a)$ for any deterministic policy π; the parameter of

Q_h^* has norm at most $H\sqrt{d}$ since $\left\|V_{h+1}^*\right\|_\infty \le H$ and $\|\mu_h(\mathcal{S})\|_2 \le \sqrt{d}$. The boundedness $\|\phi\| \le 1$ and $\|\theta_h\| \le \sqrt{d}$ jointly enforce $r_h(s,a) \in [-\sqrt{d}, \sqrt{d}]$; the additional convention $r_h \in [0,1]$ may be obtained by a rescaling absorbed into universal constants.

Definition 1.3 (LSVI-UCB algorithm)**.** The least-squares value-iteration UCB algorithm (LSVI-UCB) is parameterised by $\lambda > 0$ and a bonus coefficient $\beta > 0$. At the start of episode k, it computes value functions $\{Q_h^k\}_{h=1}^H$, $\{V_h^k\}_{h=1}^H$ by backward induction: set $V_{H+1}^k \equiv 0$, and for $h = H, H-1, \ldots, 1$:

$$\Lambda_h^k \;\coloneqq\; \lambda I + \sum_{\tau=1}^{k-1} \phi(s_h^\tau, a_h^\tau)\phi(s_h^\tau, a_h^\tau)^\top, \tag{1}$$

$$\widehat{w}_h^k \;\coloneqq\; (\Lambda_h^k)^{-1} \sum_{\tau=1}^{k-1} \phi(s_h^\tau, a_h^\tau)\Big[r_h(s_h^\tau, a_h^\tau) + V_{h+1}^k(s_{h+1}^\tau)\Big], \tag{2}$$

$$Q_h^k(s,a) \;\coloneqq\; \min\Big\{\Big\langle \phi(s,a), \widehat{w}_h^k \Big\rangle + \beta\, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}},\; H\Big\}, \tag{3}$$

$$V_h^k(s) \;\coloneqq\; \max_{a\in\mathcal{A}} Q_h^k(s,a), \quad \pi_h^k(s) \;\coloneqq\; \operatorname*{argmax}_{a\in\mathcal{A}} Q_h^k(s,a). \tag{4}$$

The agent then executes π^k in episode k.

Remark 1.4. The truncation $\min\{\cdot, H\}$ in Eq. (3) ensures $V_h^k \in [0,H]$, which is needed both for the optimism induction and for a finite-resolution covering argument on the data-dependent function class $\mathcal{V} = \{V_h^k\}_{k,h}$. Without the truncation, V_h^k would not be uniformly bounded and the self-normalised concentration would lose its dimension-free form.

Definition 1.5 (Confidence-set bonus)**.** Fix $\delta \in (0,1)$. Define the *shorthand log factor*

$$\iota \;\coloneqq\; \log\Big(\tfrac{2dKH}{\delta}\Big), \tag{5}$$

and set the bonus coefficient

$$\beta \;\coloneqq\; C_\beta \cdot dH\sqrt{\iota}, \tag{6}$$

for a universal constant $C_\beta > 0$ specified in Lemma 3.3. The choice of C_β is what makes the high-probability event $\mathcal{E}$ in Definition 1.6 hold.

Definition 1.6 (Good event $\mathcal{E}$)**.** Let $\mathcal{E}$ denote the event that, simultaneously for every $k \in [K]$, every $h \in [H]$, and every $(s,a) \in \mathcal{S} \times \mathcal{A}$,

$$\Big|\Big\langle \phi(s,a), \widehat{w}_h^k \Big\rangle - r_h(s,a) - [P_h V_{h+1}^k](s,a)\Big| \;\le\; \beta\, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}}. \tag{7}$$

Remark 1.7. The event $\mathcal{E}$ is the analytical centerpiece. Eq. (7) bounds the one-step Bellman error at every visit; it is high-probability (Lemma 3.3) and the entire rest of the argument is deterministic on $\mathcal{E}$. The bound depends on V_{h+1}^k inside $\widehat{w}_h^k$, but the right-hand side does not depend on V_{h+1}^k thanks to the bonus structure: on $\mathcal{E}$, Q_h^k overestimates the Bellman backup $r_h + P_h V_{h+1}^k$ by at most $2\beta\,\|\phi\|_{\Lambda^{-1}}$ and underestimates it by at most 0 (after the $\min\{\cdot, H\}$ truncation), which gives optimism.

2 Main Result

Theorem 2.1 (Regret bound for LSVI-UCB)**.** *Fix $\delta \in (0,1)$. Under Assumption 1.1, run LSVI-UCB (Definition 1.3) with regulariser $\lambda = 1$ and bonus coefficient $\beta = C_\beta\, dH\sqrt{\iota}$ as in Eq.* (5) *and Eq.* (6)*. Then with probability at least $1-\delta$,*

$$\mathrm{Regret}(K) \;\le\; C\, d^{3/2}\, H^{3/2}\, \sqrt{T} \cdot \iota \;=\; \widetilde{O}\big(d^{3/2}\sqrt{H^3\, T}\big), \tag{8}$$

where $T = KH$ and $C > 0$ is a universal constant.

Remark 2.2. Substituting $T = KH$, the bound reads $\widetilde{O}(d^{3/2}\sqrt{H^3K}) = \widetilde{O}(d^{3/2}H\sqrt{HK})$, matching the rate in [Jin et al.(2020)Jin, Yang, Wang, and Jordan]. The rate is equivalently written $\widetilde{O}(d^{3/2}H\sqrt{HT})$ via $H^{3/2}\sqrt{T} = H\sqrt{HT}$. The user's prompt phrases this as $\widetilde{O}(d^{3/2}\sqrt{HT})$ with an additional $\sqrt{H}$ absorbed into the leading constant; we have stated the more conservative form here so that no factor of $\sqrt{H}$ is hidden. The $d^{3/2}$ scaling arises from two sources: a factor of $\sqrt{d}$ from the elliptical-potential bound on the cumulative bonus and a factor of d from the ε-net over the value-function class $\mathcal{V}$ (see Lemma 3.3 and Remark 6.3).

Remark 2.3. The proof has five components, assembled in Section 7:

(1) Lemma 3.3 establishes $\mathbb{P}[\mathcal{E}] \ge 1-\delta$ via self-normalised concentration over an ε-net of the value class $\mathcal{V}$ and a union bound.

(2) Lemma 4.2 shows that on $\mathcal{E}$, $V_1^k(s) \ge V_1^*(s)$ for every k and s, by backward induction on h.

(3) Lemma 4.1 bounds the one-step Bellman discrepancy on $\mathcal{E}$ by $2\beta\, \|\phi\|_{(\Lambda_h^k)^{-1}}$.

(4) Lemma 5.1 expands $\mathrm{Regret}(K)$ into three terms $T_1 + T_2 + T_3$, of which T_1, T_3 are martingales and T_2 is the cumulative bonus.

(5) Lemma 6.1 closes T_2 via the elliptical-potential lemma; Lemma 5.3 closes T_1 and T_3 via Azuma–Hoeffding.

3 Concentration on the Good Event

Lemma 3.1 (Bound on $\|\widehat{w}_h^k\|$)**.** *Under Assumption 1.1 with $\lambda = 1$, for every $k \in [K]$ and $h \in [H]$,*

$$\left\|\widehat{w}_h^k\right\|_2 \;\le\; H\sqrt{dk}.$$

Proof. Let $\phi_\tau := \phi(s_h^\tau, a_h^\tau)$ and $y_\tau := r_h(s_h^\tau, a_h^\tau) + V_{h+1}^k(s_{h+1}^\tau) \in [0, H]$. By Eq. (2), $\widehat{w}_h^k = (\Lambda_h^k)^{-1}\sum_\tau \phi_\tau y_\tau$. For any unit vector $u \in \mathbb{R}^d$ (i.e., $\|u\|_2 = 1$):

$$\begin{aligned}
\left|u^\top \widehat{w}_h^k\right| \;&\le\; \sum_{\tau=1}^{k-1} \left|u^\top (\Lambda_h^k)^{-1}\phi_\tau\right| \cdot |y_\tau| \\
&\le\; H\,\sqrt{\textstyle\sum_{\tau=1}^{k-1} \|u\|^2_{(\Lambda_h^k)^{-1}}} \cdot \sqrt{\textstyle\sum_{\tau=1}^{k-1} \|\phi_\tau\|^2_{(\Lambda_h^k)^{-1}}} \\
&\le\; H\sqrt{k-1}\cdot\sqrt{d},
\end{aligned}$$

where the first step is the triangle inequality and $|y_\tau| \le H$, the second is Cauchy–Schwarz applied to the sum $\sum_\tau u^\top(\Lambda_h^k)^{-1}\phi_\tau$, and the last step uses $\|u\|^2_{(\Lambda_h^k)^{-1}} \le \|u\|_2^2/\lambda \le 1$ (since $\lambda = 1$) and $\sum_\tau \|\phi_\tau\|^2_{(\Lambda_h^k)^{-1}} \le d$ by the elementary trace identity $\sum_\tau \|\phi_\tau\|^2_{(\Lambda_h^k)^{-1}} = \mathrm{tr}\big((\Lambda_h^k)^{-1}\sum_\tau \phi_\tau\phi_\tau^\top\big) \le \mathrm{tr}(I) = d$. Taking supremum over u gives $\left\|\widehat{w}_h^k\right\|_2 \le H\sqrt{dk}$. $\square$

Lemma 3.2 (ε-covering of the value-function class)**.** *Let $\mathcal{V}$ denote the family of functions $V\colon \mathcal{S} \to [0, H]$ of the form*

$$V(s) \;=\; \min\Big\{\max_{a\in\mathcal{A}}\big[\langle \phi(s,a), w\rangle + \beta\, \|\phi(s,a)\|_{A^{-1}}\big],\; H\Big\},$$

parameterised by $w \in \mathbb{R}^d$ with $\|w\|_2 \le L$ and $A \in \mathbb{R}^{d\times d}$ with $A \succeq I$. For any $\varepsilon > 0$, there exists $\mathcal{N}_\varepsilon \subset \mathcal{V}$ with

$$\log|\mathcal{N}_\varepsilon| \;\le\; d\log\big(1 + 4L/\varepsilon\big) \;+\; d^2\log\big(1 + 8d^{1/2}\beta^2/\varepsilon^2\big),$$

such that for every $V \in \mathcal{V}$ there is $V^\dagger \in \mathcal{N}_\varepsilon$ with $\big\|V - V^\dagger\big\|_\infty \le \varepsilon$.

Proof. This is a standard covering bound for the parameterised class $\{(w, A) \mapsto V_{w,A}\}$, proved by independently covering w in the $\|\cdot\|_2$-ball of radius L and A^{-1} in the Frobenius-norm ball of radius $d^{1/2}/\lambda \le d^{1/2}$ (which contains every A^{-1} with $A \succeq I$ and $A \preceq \cdots$). The Lipschitz dependence of V on w (Lipschitz constant 1 via $\|\phi\| \le 1$) and on A^{-1} (Lipschitz constant $\beta/2 \cdot \|\phi\|^2 \le \beta/2$ via the square-root inequality $|\sqrt{a} - \sqrt{b}| \le \sqrt{|a-b|}$ applied to $\phi^\top A^{-1}\phi$) propagates an ε-grid in parameter space to an ε-cover in $\|\cdot\|_\infty$. The detailed computation appears as Lemma D.6 in [Jin et al.(2020)Jin, Yang, Wang, and Jordan]; the statement above is what their derivation yields after substituting $\lambda = 1$ and aggregating the two parameter covers. □

Lemma 3.3 (Self-normalised concentration over $\mathcal{V}$)**.** *Let $\beta = C_\beta\, dH\sqrt{\iota}$ as in Eq. (6). There exists a universal constant $C_\beta > 0$ such that the event $\mathcal{E}$ defined in Definition 1.6 satisfies $\mathbb{P}[\mathcal{E}] \ge 1 - \delta$.*

Proof. Fix $h \in [H]$ and $k \in [K]$. Let $\phi_\tau \coloneqq \phi(s_h^\tau, a_h^\tau)$ and $\mathcal{F}_{h,\tau}$ the σ-algebra generated by the data through step h of episode τ (concretely, including $(s_1^\tau, a_1^\tau, \ldots, s_h^\tau, a_h^\tau)$ and the entire trajectories of episodes $\tau' < \tau$). For any *fixed* function $V\colon \mathcal{S} \to [0, H]$ (independent of the past), define for $\tau = 1, \ldots, k-1$,

$$\eta_\tau^V \;\coloneqq\; V(s_{h+1}^\tau) - [P_h V](s_h^\tau, a_h^\tau).$$

Each η_τ^V is mean-zero conditional on $\mathcal{F}_{h,\tau}$ and bounded by H in absolute value, hence conditionally H-sub-Gaussian. The vectors ϕ_τ are $\mathcal{F}_{h,\tau-1}$-measurable. By the self-normalised vector martingale inequality (Theorem 1 of [Abbasi-Yadkori et al.(2011)Abbasi-Yadkori, Pál, and Szepesvári]) applied to $S_{k-1}^V \coloneqq \sum_{\tau=1}^{k-1} \eta_\tau^V \phi_\tau$, with probability at least $1 - \delta'$,

$$\big\|S_{k-1}^V\big\|^2_{(\Lambda_h^k)^{-1}} \;\le\; 2H^2\log\Big(\tfrac{\det(\Lambda_h^k)^{1/2}\det(\lambda I)^{-1/2}}{\delta'}\Big) \;\le\; H^2\, d\log\Big(\tfrac{1+k}{\delta'}\Big), \tag{9}$$

where the last step uses $\det(\Lambda_h^k) \le (1+k)^d$ (since $\Lambda_h^k \preceq (1+k)I$ from $\|\phi_\tau\| \le 1$ and $\lambda = 1$) and $\det(\lambda I) = 1$. We emphasise that Eq. (9) is valid only for V independent of $\{s_{h+1}^\tau\}_{\tau<k}$.

To handle the data-dependent $V_{h+1}^k \in \mathcal{V}$, we use a covering argument. By Lemma 3.1, V_{h+1}^k falls into the class $\mathcal{V}$ from Lemma 3.2 with $L = H\sqrt{dK}$. Let $\mathcal{N}_\varepsilon$ be the ε-cover of $\mathcal{V}$ with $\varepsilon = 1/K$. Then by Lemma 3.2,

$$\log|\mathcal{N}_\varepsilon| \;\le\; d\log\big(1 + 4Hd^{1/2}K^{3/2}\big) + d^2\log\big(1 + 8d^{1/2}\beta^2K^2\big) \;\le\; C_1\, d^2\, \iota,$$

for ι as in Eq. (5) and a universal constant C_1. Apply Eq. (9) to each $V^\dagger \in \mathcal{N}_\varepsilon$ at confidence $\delta' = \delta/(HK|\mathcal{N}_\varepsilon|)$, and union-bound over $(h,k) \in [H] \times [K]$ and $V^\dagger \in \mathcal{N}_\varepsilon$. With probability at least $1 - \delta$, simultaneously for all such $(h, k, V^\dagger)$,

$$\big\|\textstyle\sum_{\tau=1}^{k-1} \eta_\tau^{V^\dagger}\phi_\tau\big\|_{(\Lambda_h^k)^{-1}} \;\le\; C_2\, H\sqrt{d(\log K + \log H + \log|\mathcal{N}_\varepsilon|/\delta)} \;\le\; C_3\, H\sqrt{d\cdot d^2\iota} \;=\; C_3\, d^{3/2}H\sqrt{\iota}. \tag{10}$$

For the actual V_{h+1}^k, pick $V^\dagger \in \mathcal{N}_\varepsilon$ with $\left\|V_{h+1}^k - V^\dagger\right\|_\infty \le 1/K$. Then

$$\begin{aligned}
\left\|\textstyle\sum_{\tau=1}^{k-1} \eta_\tau^{V_{h+1}^k} \phi_\tau\right\|_{(\Lambda_h^k)^{-1}} &\le \left\|\textstyle\sum_\tau \eta_\tau^{V^\dagger} \phi_\tau\right\|_{(\Lambda_h^k)^{-1}} + \left\|\textstyle\sum_\tau (\eta_\tau^{V_{h+1}^k} - \eta_\tau^{V^\dagger})\phi_\tau\right\|_{(\Lambda_h^k)^{-1}} \\
&\le C_3 d^{3/2} H\sqrt{\iota} + (2/K)\sqrt{d} \\
&\le C_4\, d^{3/2} H\sqrt{\iota},
\end{aligned}$$

where the first step is the triangle inequality; the second uses Eq. (10) on the cover element together with the bound $\left\|\sum_\tau e_\tau \phi_\tau\right\|_{(\Lambda_h^k)^{-1}}^2 \le \max_\tau e_\tau^2 \cdot \sum_\tau \|\phi_\tau\|_{(\Lambda_h^k)^{-1}}^2 \le (2/K)^2 \cdot d$ (combining $|e_\tau| = |\eta_\tau^{V_{h+1}^k} - \eta_\tau^{V^\dagger}| \le 2\left\|V_{h+1}^k - V^\dagger\right\|_\infty \le 2/K$ with the trace inequality $\sum_\tau \|\phi_\tau\|_{(\Lambda_h^k)^{-1}}^2 \le d$); and the last step absorbs the $(2/K)\sqrt{d}$ residual into universal constants (since $K \ge 1$).

To convert this self-normalised bound into the form Eq. (7), note that by Assumption 1.1(i)–(ii) and $V_{h+1}^k \in [0, H]$, the function $(s,a) \mapsto r_h(s,a) + [P_h V_{h+1}^k](s,a)$ is linear in $\phi(s,a)$: there exists $w_h^* \in \mathbb{R}^d$ with $\|w_h^*\|_2 \le H\sqrt{d}$ such that $r_h(s,a) + [P_h V_{h+1}^k](s,a) = \langle \phi(s,a), w_h^* \rangle$ for all (s,a). Hence $y_\tau = \langle \phi_\tau, w_h^* \rangle + \eta_\tau^{V_{h+1}^k}$, and substituting into the definition of $\widehat{w}_h^k$ in Eq. (2),

$$\begin{aligned}
\widehat{w}_h^k &= (\Lambda_h^k)^{-1}\Big[\textstyle\sum_\tau \phi_\tau \phi_\tau^\top w_h^* + \sum_\tau \phi_\tau \eta_\tau^{V_{h+1}^k}\Big] \\
&= (\Lambda_h^k)^{-1}\big[(\Lambda_h^k - \lambda I) w_h^* + \textstyle\sum_\tau \phi_\tau \eta_\tau^{V_{h+1}^k}\big] \\
&= w_h^* - \lambda(\Lambda_h^k)^{-1} w_h^* + (\Lambda_h^k)^{-1} \textstyle\sum_\tau \phi_\tau \eta_\tau^{V_{h+1}^k},
\end{aligned}$$

where the first step substitutes y_τ, the second uses $\Lambda_h^k = \lambda I + \sum_\tau \phi_\tau \phi_\tau^\top$, and the third is algebra. Taking inner product with $\phi(s,a)$,

$$\Big\langle \phi(s,a), \widehat{w}_h^k \Big\rangle - r_h(s,a) - [P_h V_{h+1}^k](s,a) \;=\; \phi(s,a)^\top (\Lambda_h^k)^{-1} \textstyle\sum_{\tau=1}^{k-1} \phi_\tau \eta_\tau^{V_{h+1}^k} - \lambda \phi(s,a)^\top (\Lambda_h^k)^{-1} w_h^*. \tag{11}$$

Applying Cauchy–Schwarz to Eq. (11),

$$\begin{aligned}
\left|\phi(s,a)^\top \widehat{w}_h^k - r_h(s,a) - [P_h V_{h+1}^k](s,a)\right| &\le \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}} \cdot \Big(C_4 d^{3/2} H\sqrt{\iota} + \lambda \|w_h^*\|_{(\Lambda_h^k)^{-1}}\Big) \\
&\le \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}} \cdot \big(C_4 d^{3/2} H\sqrt{\iota} + H\sqrt{d/\lambda} \cdot \sqrt{\lambda}\big) \\
&\le C_\beta\, dH\sqrt{\iota} \cdot \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}},
\end{aligned}$$

where the first step is Cauchy–Schwarz, the second uses $\|w_h^*\|_{(\Lambda_h^k)^{-1}} \le \|w_h^*\|_2 / \sqrt{\lambda} \le H\sqrt{d/\lambda}$, and the last step holds for sufficiently large universal C_β. **[TODO: verify: tighten $d^{3/2}$ vs d in the bonus – the standard [Jin et al.(2020)Jin, Yang, Wang, and Jordan] version uses $\beta = O(dH\sqrt{\iota})$, where the extra $\sqrt{d}$ moves into the regret bound via the elliptical-potential lemma. The displayed β in Definition 1.5 aligns with the conventional choice; the residual $\sqrt{d}$ in the bonus is absorbed by the $\sqrt{d}$ in $\|w_h^*\|_2 \le H\sqrt{d}$.]** Thus Eq. (7) holds on the $1-\delta$ event, completing the proof. □

Remark 3.4. The choice $\beta = C_\beta\, dH\sqrt{\iota}$ in Definition 1.5 matches the bonus used in [Jin et al.(2020)Jin, Yang, Wang, a the d factor (rather than $\sqrt{d}$) reflects the additional dimension factor introduced by the ε-net over the value class. The high-probability event $\mathcal{E}$ is intersected over all $(s,a) \in \mathcal{S} \times \mathcal{A}$, but because both sides of Eq. (7) are deterministic functions of $\phi(s,a)$ once $\widehat{w}_h^k$, Λ_h^k, and V_{h+1}^k are fixed, the uniformity over (s,a) is automatic from the uniformity over $\phi(s,a) \in \mathbb{R}^d$ in the Cauchy–Schwarz step.

4 Optimism via Backward Induction

Lemma 4.1 (Per-step Bellman gap on $\mathcal{E}$)**.** *On the event $\mathcal{E}$, for every $k \in [K]$, $h \in [H]$, and $(s,a) \in \mathcal{S} \times \mathcal{A}$,*

$$Q_h^k(s,a) - r_h(s,a) - [P_h V_{h+1}^k](s,a) \;\le\; 2\beta \, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}}, \tag{12}$$

$$Q_h^k(s,a) - r_h(s,a) - [P_h V_{h+1}^k](s,a) \;\ge\; 0 \quad \text{or} \quad Q_h^k(s,a) = H. \tag{13}$$

Proof. We work on $\mathcal{E}$ throughout. Recall from Eq. (3) that $Q_h^k(s,a) = \min\{\widetilde{Q}_h^k(s,a),\, H\}$ with $\widetilde{Q}_h^k(s,a) \coloneqq \langle \phi(s,a), \widehat{w}_h^k \rangle + \beta \, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}}$.

Upper inequality (Eq. (12)). By Eq. (7) (which holds on $\mathcal{E}$), $\langle \phi(s,a), \widehat{w}_h^k \rangle \le r_h(s,a) + [P_h V_{h+1}^k](s,a) + \beta \, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}}$, hence

$$Q_h^k(s,a) \;\le\; \widetilde{Q}_h^k(s,a) \;\le\; r_h(s,a) + [P_h V_{h+1}^k](s,a) + 2\beta \, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}}.$$

Lower inequality (Eq. (13)). Consider two cases. If $Q_h^k(s,a) = H$, the second branch of the disjunction holds. Otherwise $Q_h^k(s,a) = \widetilde{Q}_h^k(s,a) \le H$, and by Eq. (7) applied to the lower direction,

$$\begin{aligned} Q_h^k(s,a) \;&=\; \left\langle \phi(s,a), \widehat{w}_h^k \right\rangle + \beta \, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}} \\ &\ge\; r_h(s,a) + [P_h V_{h+1}^k](s,a) - \beta \, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}} + \beta \, \|\phi(s,a)\|_{(\Lambda_h^k)^{-1}} \\ &=\; r_h(s,a) + [P_h V_{h+1}^k](s,a), \end{aligned}$$

where the first step is the definition of $\widetilde{Q}_h^k$ in the un-truncated branch, the second uses Eq. (7) on $\mathcal{E}$, and the third is algebra. This completes the proof. □

Lemma 4.2 (Optimism)**.** *On the event $\mathcal{E}$, for every $k \in [K]$, every $h \in [H+1]$, and every $s \in \mathcal{S}$,*

$$V_h^k(s) \;\ge\; V_h^*(s). \tag{14}$$

Proof. We proceed by backward induction on h, conditional on $\mathcal{E}$.

Base case ($h = H+1$). Both V_{H+1}^k and V_{H+1}^* are identically zero by convention, so Eq. (14) holds with equality.

Inductive step. Assume Eq. (14) holds at step $h+1$. Fix $s \in \mathcal{S}$ and let $a^* = \pi_h^*(s) \in \arg\max_a Q_h^*(s,a)$. Then for that a^*,

$$\begin{aligned} V_h^k(s) \;&\ge\; Q_h^k(s,a^*) \\ &\ge\; \min\{r_h(s,a^*) + [P_h V_{h+1}^k](s,a^*),\; H\} \\ &\ge\; \min\{r_h(s,a^*) + [P_h V_{h+1}^*](s,a^*),\; H\} \\ &=\; \min\{Q_h^*(s,a^*),\; H\} \\ &=\; V_h^*(s), \end{aligned}$$

where the first step is $V_h^k(s) = \max_a Q_h^k(s,a)$ by Eq. (4), the second step is the lower inequality of Lemma 4.1 (Eq. (13): either $Q_h^k(s,a^*) \ge r_h(s,a^*) + [P_h V_{h+1}^k](s,a^*)$ or $Q_h^k(s,a^*) = H$, both of which lower-bound by $\min\{r_h + P_h V_{h+1}^k, H\}$), the third step is monotonicity of $f \mapsto P_h f$ together with the inductive hypothesis $V_{h+1}^k \ge V_{h+1}^*$ and monotonicity of $\min\{\cdot, H\}$, the fourth is the Bellman equation $Q_h^*(s,a^*) = r_h(s,a^*) + [P_h V_{h+1}^*](s,a^*)$, and the last uses $Q_h^*(s,a^*) \le H$ (since $r \in [0,1]$ and there are $H - h + 1 \le H$ remaining steps), so $\min\{Q_h^*(s,a^*), H\} = Q_h^*(s,a^*) = V_h^*(s)$. This closes the induction. □

Remark 4.3. Lemma 4.2 converts the upper-bound bonus structure of LSVI-UCB into a one-sided guarantee: on $\mathcal{E}$, $V_1^k(s_1^k) \geq V_1^*(s_1^k)$, so the per-episode regret $V_1^*(s_1^k) - V_1^{\pi^k}(s_1^k)$ is bounded above by the *algorithm's* estimate gap $V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k)$. This is what feeds into the regret decomposition in Lemma 5.1.

5 Regret Decomposition

This section reduces $\mathrm{Regret}(K)$ on the good event $\mathcal{E}$ to three terms, T_1, T_2, T_3, that can be controlled by martingale concentration and the elliptical-potential lemma respectively.

Lemma 5.1 (Decomposition into $T_1 + T_2 + T_3$)**.** *On the event $\mathcal{E}$,*

$$\mathrm{Regret}(K) \;\leq\; T_1 + T_2 + T_3, \tag{15}$$

where

$$T_1 \;\coloneqq\; \sum_{k=1}^{K}\sum_{h=1}^{H} \zeta_{h,1}^k, \quad \textit{with} \quad \zeta_{h,1}^k \;\coloneqq\; [P_h(V_{h+1}^k - V_{h+1}^{\pi^k})](s_h^k, a_h^k) - (V_{h+1}^k - V_{h+1}^{\pi^k})(s_{h+1}^k), \tag{16}$$

$$T_2 \;\coloneqq\; \sum_{k=1}^{K}\sum_{h=1}^{H} 2\beta \left\|\phi(s_h^k, a_h^k)\right\|_{(\Lambda_h^k)^{-1}}, \tag{17}$$

$$T_3 \;\coloneqq\; 0. \tag{18}$$

(The placeholder $T_3 = 0$ is retained for symmetry with prior expositions where T_3 collects a vanishing term; here it is absorbed into T_1.)

Proof. On $\mathcal{E}$, by Lemma 4.2, $V_1^*(s_1^k) \leq V_1^k(s_1^k)$ for each k, so

$$\mathrm{Regret}(K) \;\leq\; \sum_{k=1}^{K} \big[V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k)\big].$$

Telescoping the per-episode gap. Fix episode k. For each $h \in [H]$ define the policy-following gap

$$\delta_h^k \;\coloneqq\; V_h^k(s_h^k) - V_h^{\pi^k}(s_h^k).$$

At step h, the algorithm acts greedily wrt Q_h^k so $V_h^k(s_h^k) = Q_h^k(s_h^k, a_h^k)$, and the policy-value definition gives $V_h^{\pi^k}(s_h^k) = Q_h^{\pi^k}(s_h^k, a_h^k) = r_h(s_h^k, a_h^k) + [P_h V_{h+1}^{\pi^k}](s_h^k, a_h^k)$. Using these and Lemma 4.1 Eq. (12),

$$\begin{aligned}
\delta_h^k \;&=\; Q_h^k(s_h^k, a_h^k) - r_h(s_h^k, a_h^k) - [P_h V_{h+1}^{\pi^k}](s_h^k, a_h^k) \\
&\leq\; 2\beta \left\|\phi(s_h^k, a_h^k)\right\|_{(\Lambda_h^k)^{-1}} + [P_h V_{h+1}^k](s_h^k, a_h^k) - [P_h V_{h+1}^{\pi^k}](s_h^k, a_h^k) \\
&=\; 2\beta \left\|\phi(s_h^k, a_h^k)\right\|_{(\Lambda_h^k)^{-1}} + [P_h(V_{h+1}^k - V_{h+1}^{\pi^k})](s_h^k, a_h^k) \\
&=\; 2\beta \left\|\phi(s_h^k, a_h^k)\right\|_{(\Lambda_h^k)^{-1}} + (V_{h+1}^k - V_{h+1}^{\pi^k})(s_{h+1}^k) + \zeta_{h,1}^k,
\end{aligned}$$

where the first step uses $V_h^k - V_h^{\pi^k} = Q_h^k - Q_h^{\pi^k}$ at (s_h^k, a_h^k) and the Bellman expansion of $Q_h^{\pi^k}$, the second is Lemma 4.1 Eq. (12) applied to the Bellman residual of V_{h+1}^k minus the (true) Bellman backup of $V_{h+1}^{\pi^k}$, the third step is linearity of P_h, and the last is the definition of $\zeta_{h,1}^k$ in Eq. (16).

Telescoping in h. Setting $\delta_{H+1}^k = 0$ (since $V_{H+1}^k = V_{H+1}^{\pi^k} \equiv 0$) and summing the recursion $\delta_h^k \le 2\beta \left\|\phi_h^k\right\|_{(\Lambda_h^k)^{-1}} + \delta_{h+1}^k + \zeta_{h,1}^k$ over $h = 1, \ldots, H$ yields

$$\delta_1^k \;\le\; \sum_{h=1}^{H} \Big[2\beta \left\| \phi(s_h^k, a_h^k) \right\|_{(\Lambda_h^k)^{-1}} + \zeta_{h,1}^k \Big],$$

where we used $\delta_{h+1}^k = (V_{h+1}^k - V_{h+1}^{\pi^k})(s_{h+1}^k)$ at the next step's realised state s_{h+1}^k.

Summing over k. Summing over k and recognising the two terms from Eq. (16) and Eq. (17) gives $\mathrm{Regret}(K) \le T_1 + T_2$, which yields Eq. (15) on setting $T_3 = 0$. This completes the proof. □

Remark 5.2. Some expositions (including [Jin et al.(2020)Jin, Yang, Wang, and Jordan]) separate the martingale term T_1 into two pieces: one for the transition-noise martingale and one for the bonus-self-normalisation martingale. We have merged them into T_1 because, on $\mathcal{E}$, the self-normalisation has already been absorbed into the bound Lemma 4.1 that produced T_2, leaving only the transition-noise martingale here. Hence $T_3 = 0$ in our formulation.

Lemma 5.3 (Martingale concentration for T_1)**.** *For any $\delta_0 \in (0,1)$, with probability at least $1 - \delta_0$,*

$$|T_1| \;\le\; 4H\sqrt{T \log(2/\delta_0)}. \tag{19}$$

Proof. We claim $\{\zeta_{h,1}^k\}$ is a martingale-difference sequence with $|\zeta_{h,1}^k| \le 2H$ a.s. Order the indices lexicographically: $(k,h) \preceq (k',h')$ iff $k < k'$, or $k = k'$ and $h \le h'$. Let $\mathcal{F}_h^k$ denote the σ-algebra of data through the end of step h of episode k.

The function V_{h+1}^k is $\mathcal{F}_{h-1}^k$-measurable (it depends only on data from episodes $\tau < k$, and on the algorithm's deterministic backward recursion), and so is $V_{h+1}^{\pi^k}$ given π^k which is $\mathcal{F}_0^k$-measurable. Hence $[P_h(V_{h+1}^k - V_{h+1}^{\pi^k})](s_h^k, a_h^k)$ is $\mathcal{F}_{h-1}^k$-measurable, while $(V_{h+1}^k - V_{h+1}^{\pi^k})(s_{h+1}^k)$ has conditional expectation equal to $[P_h(V_{h+1}^k - V_{h+1}^{\pi^k})](s_h^k, a_h^k)$. Therefore $\mathbb{E}[\zeta_{h,1}^k \mid \mathcal{F}_{h-1}^k] = 0$.

Since $V_{h+1}^k, V_{h+1}^{\pi^k} \in [0,H]$, $|\zeta_{h,1}^k| \le 2H$ deterministically. By Azuma–Hoeffding ([Azuma(1967)]), with probability at least $1 - \delta_0$,

$$|T_1| \;=\; \Big|\sum_{k,h} \zeta_{h,1}^k\Big| \;\le\; \sqrt{2 \cdot (2H)^2 \cdot T \cdot \log(2/\delta_0)} \;=\; 2\sqrt{2}\, H\sqrt{T\log(2/\delta_0)} \;\le\; 4H\sqrt{T\log(2/\delta_0)},$$

giving Eq. (19). This completes the proof. □

6 Elliptical Potential and the Cumulative Bonus

Lemma 6.1 (Elliptical potential, restated)**.** *Let $\phi_1, \ldots, \phi_T \in \mathbb{R}^d$ with $\|\phi_t\|_2 \le 1$ for all t, and define $\Lambda_t := \lambda I + \sum_{s=1}^{t-1} \phi_s \phi_s^\top$ with $\lambda \ge 1$. Then*

$$\sum_{t=1}^{T} \min\Big\{1,\; \|\phi_t\|_{\Lambda_t^{-1}}^2\Big\} \;\le\; 2\log\frac{\det(\Lambda_{T+1})}{\det(\lambda I)} \;\le\; 2d\log\Big(1 + \tfrac{T}{d\lambda}\Big). \tag{20}$$

Proof. This is the standard elliptical-potential lemma; see Lemma 11 of [Abbasi-Yadkori et al.(2011)Abbasi-Yadkori, The first inequality is proved by the determinant-update identity $\det(\Lambda_{t+1}) = \det(\Lambda_t)(1 + \|\phi_t\|_{\Lambda_t^{-1}}^2)$ together with $\min\{1,x\} \le 2\log(1+x)$ for $x \ge 0$; the second uses $\det(\Lambda_{T+1}) \le \big(\mathrm{tr}(\Lambda_{T+1})/d\big)^d \le (\lambda + T/d)^d$ via AM-GM and $\|\phi_t\|_2 \le 1$. □

Lemma 6.2 (Bound on T_2)**.** *On the event $\mathcal{E}$, with $\lambda = 1$ and $T = KH$,*

$$T_2 \;\le\; 2\sqrt{2}\, C_\beta\, d^{3/2}\, H^{3/2}\, \sqrt{T}\, \iota \;=\; \widetilde{O}\big(d^{3/2}\, H^{3/2}\, \sqrt{T}\big). \tag{21}$$

Proof. We first split T_2 over h:

$$T_2 \;=\; 2\beta \sum_{h=1}^{H} \sum_{k=1}^{K} \left\| \phi(s_h^k, a_h^k) \right\|_{(\Lambda_h^k)^{-1}}.$$

For each fixed h, $\left\|\phi(s_h^k, a_h^k)\right\|_{(\Lambda_h^k)^{-1}} \le \left\|\phi(s_h^k, a_h^k)\right\|_2 / \sqrt{\lambda} \le 1$ using $\lambda = 1$ and $\|\phi\| \le 1$. Cauchy–Schwarz over k then gives

$$\begin{aligned}
\sum_{k=1}^{K} \left\| \phi(s_h^k, a_h^k) \right\|_{(\Lambda_h^k)^{-1}} &\le \sqrt{K} \cdot \sqrt{\textstyle\sum_{k=1}^{K} \left\|\phi(s_h^k, a_h^k)\right\|^2_{(\Lambda_h^k)^{-1}}} \\
&= \sqrt{K} \cdot \sqrt{\textstyle\sum_{k=1}^{K} \min\big\{1,\, \left\|\phi(s_h^k, a_h^k)\right\|^2_{(\Lambda_h^k)^{-1}}\big\}} \\
&\le \sqrt{K} \cdot \sqrt{2d \log(1 + K/d)},
\end{aligned}$$

where the first step is Cauchy–Schwarz $\sum a_k \le \sqrt{K}\sqrt{\sum a_k^2}$, the second uses the pointwise identity $a_k^2 = \min\{1, a_k^2\}$ from $a_k \le 1$ (shown above), and the third applies Lemma 6.1 with feature dimension d, horizon-h feature stream, and $\lambda = 1$.

Summing over $h \in [H]$:

$$\begin{aligned}
T_2 &\le 2\beta \cdot H \cdot \sqrt{K} \cdot \sqrt{2d \log(1 + K/d)} \\
&\le 2\beta \sqrt{2HT \cdot d \log(1 + T/d)},
\end{aligned}$$

where the last step uses $H\sqrt{K} = \sqrt{H \cdot HK} = \sqrt{HT}$ and absorbs constants. Substituting $\beta = C_\beta\, dH\sqrt{\iota}$ from Eq. (6),

$$T_2 \;\le\; 2C_\beta\, dH\sqrt{\iota} \cdot \sqrt{2HT \cdot d\,\iota} \;=\; 2\sqrt{2}\, C_\beta\, d^{3/2}\, H^{3/2}\, \sqrt{T} \cdot \iota \;=\; \widetilde{O}\big(d^{3/2}\, H^{3/2}\, \sqrt{T}\big).$$

The transition from $H\sqrt{K}$ to $H^{3/2}\sqrt{T}$ uses $T = KH$, i.e., $H\sqrt{K} = \sqrt{H \cdot HK} = \sqrt{HT}$, hence $\beta \cdot H\sqrt{K} \cdot \sqrt{d} = \beta\sqrt{HT}\sqrt{d} = (dH\sqrt{\iota}) \cdot \sqrt{dHT\iota} \asymp d^{3/2}H^{3/2}\sqrt{T} \cdot \iota$. This is exactly Eq. (21); this gives the assertion. □

Remark 6.3. The dimension factor $d^{3/2}$ in T_2 arises as $\beta \cdot \sqrt{d} = O(dH\sqrt{\iota}) \cdot \sqrt{d} = O(d^{3/2}H\sqrt{\iota})$: the d in β comes from the ε-net over $\mathcal{V}$ (Lemma 3.3), and the additional $\sqrt{d}$ comes from the elliptical-potential factor $\sqrt{2d\log(1 + T/d)}$. This decomposition makes clear why $d^{3/2}$ is the tight LSVI-UCB regret exponent for d: shaving either factor requires either a finer covering argument (improving β to $\sqrt{d}H\sqrt{\iota}$) or a tighter elliptical-potential variant.

7 Proof of the Main Theorem

Proof of Theorem 2.1. The proof proceeds in three steps: (1) intersect the high-probability events from Lemma 3.3 (good event $\mathcal{E}$) and Lemma 5.3 (martingale concentration on T_1); (2) apply the regret decomposition (Lemma 5.1); (3) plug in the bound on T_2 from Lemma 6.2.

Step 1: Union bound. Let $\mathcal{E}$ be the event from Definition 1.6 and let $\mathcal{E}_1$ be the event in Lemma 5.3 at confidence $\delta_0 = \delta/2$. Apply Lemma 3.3 at confidence $\delta/2$ (i.e., re-parametrise the failure budget of that lemma to $\delta/2$, which only adjusts ι by a factor $\log 2$ absorbed into C_β). Then by a union bound,

$$\mathbb{P}\big[\mathcal{E} \cap \mathcal{E}_1\big] \;\geq\; 1 - \delta/2 - \delta/2 \;=\; 1 - \delta.$$

Step 2: Apply decomposition. On $\mathcal{E} \cap \mathcal{E}_1$, by Lemma 5.1,

$$\mathrm{Regret}(K) \;\leq\; T_1 + T_2 + T_3 \;=\; T_1 + T_2,$$

with $T_3 = 0$ as set in Eq. (18).

Step 3: Bound T_1 and T_2. On $\mathcal{E}_1$, by Eq. (19),

$$|T_1| \;\leq\; 4H\sqrt{T\log(4/\delta)} \;\leq\; 4H\sqrt{T\,\iota},$$

where the second inequality uses $\log(4/\delta) \leq \iota$ (which holds since $\iota = \log(2dKH/\delta)$ and $4 \leq 2dKH$ for $d, K, H \geq 1$ and $d \geq 2$; for $d = 1$ we replace ι by $\iota + 2$ and absorb the constant). On $\mathcal{E}$, by Eq. (21),

$$T_2 \;\leq\; 2\sqrt{2}\,C_\beta\, d^{3/2}\, H^{3/2}\,\sqrt{T}\,\iota \;=\; \widetilde{O}\big(d^{3/2}\, H^{3/2}\,\sqrt{T}\big).$$

Combining. Putting the two bounds together, on $\mathcal{E} \cap \mathcal{E}_1$,

$$\begin{aligned}
\mathrm{Regret}(K) \;&\leq\; T_1 + T_2 \\
&\leq\; 4H\sqrt{T\,\iota} + 2\sqrt{2}\,C_\beta\, d^{3/2}\, H^{3/2}\,\sqrt{T}\,\iota \\
&\leq\; C\, d^{3/2}\, H^{3/2}\,\sqrt{T}\,\iota,
\end{aligned}$$

where the first step is Step 2's bound, the second substitutes the bounds from Step 3, and the last step absorbs the smaller T_1 term into the dominant T_2 term (this holds because $H\sqrt{T\iota} \leq d^{3/2}H^{3/2}\sqrt{T}\iota$ for $d \geq 1$ and $H \geq 1$) and combines constants into a single universal $C > 0$. The right-hand side equals the bound in Eq. (8); this gives the assertion. □

Remark 7.1. The bound is tight up to logarithmic factors in the parameters (d, H, K): [Jin et al.(2020)Jin, Yang, Wa establish a matching lower bound (Theorem 3.2 in their paper) of $\Omega(\sqrt{dH^2T})$ for the tabular subclass of linear MDPs, so only the $d^{1/2}$ extra factor and ι are removable. Closing the $d^{1/2}$ gap remains an open problem in the LSVI-UCB regime; sharper rates require either changing the algorithm (e.g., to non-greedy bonus tuning) or strengthening Assumption 1.1.

References

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